# Title: Behavioural Threat Mitigation & Detection (Pyramid of Pain Case)

## Executive Summary

Analyze an evolving intrusion campaign and deploying counter measures. A malicious execution file was used.

## Static Indicators

The malicious file was scanned using Malware Sandbox. Blocking the SHA-256, bocking the egress IP addresses and filtering the DNS did not stop the evolving attack.

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|---|---|
| Add SHA-256 Hash to blocklist | Hash will be different even with the smallest change in the file. |
| Blocking IP Address | Attacker uses VPN to change IP address. |
| Filter Out DNS | Attacker purchase and register new domain name. |

## Behavioral Detection

The malware evolved to modify registry to evade defense. Sigma rules were used to identify the registry activity.

The attacker changed the attack method so that the instruction occurs on his back-end server. Using Sigma, Network Connections rules were set according to the observation (file size and frequency) of artifacts in Network Activity of the user.

Finally, the attacker uses remote access to attack. It executes cmd.exe and ran his commands. After reviewing the command log, Sigma rule was set to stop the cmd.exe process that contains specific string used by the attacker.dd

## MITRE ATT&CK

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|---|---|
| Executing malicious file | T1204.002 \| User Execution: Malicious File |
| HTTP Request | T1071.001 \| Application Layer Protocol: Web Protocols |
| VPN used to evade IP filtering | T1071.004 \| Application Layer Protocol: DNS |
| Changes Registry to impair defense | T1112 \| Modify Registry |
| Conseutive HTTP POST requests | T1102.002 \| Web Service: Bidirectional Communication |
| Remote Access and execute CMD.exe | T1059.003 \| Command & Scripting Interpreter: Windows Command Shell |